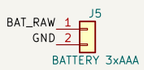
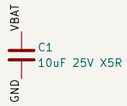
Button-activated Soil Moisture Meter – 3xAAA (3.0–4.8V), no regulator
Idle: MCU in power-down, TPS22917 off –> ~0.5 uA total. Press SW1 –> PA6 wakes MCU –>
LOAD_EN high –> VSW powers OLED + sensor –> ADC reads AOUT (PA5) –> display 15 s –> LOAD_EN low –> sleep.



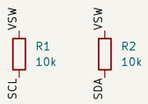


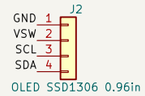




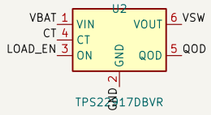


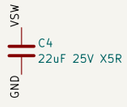






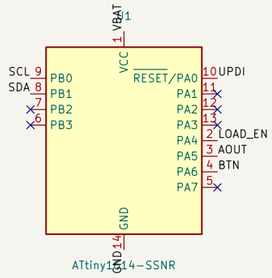




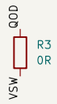




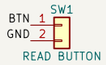












Firmware notes: run at 10 MHz (valid >2.7 V). Before LOAD_EN low, disable TWI and set
PB0/PB1 to input w/o pull-up so the unpowered OLED/sensor aren't back-fed.
Sensor read ratiometric (ADC ref = VDD, sensor LDO bypassed). Battery: VDD vs. internal reference.
OLED module runs from VSW via its own LDO (3.3–5V). Low-batt cutoff ~3.3 V (1.1 V/cell).

2xAAA, ATtiny1614, TPS22917 load switch, SSD1306 OLED, STEMMA soil sensor

Sheet: /
File: Moisture_Sensor.kicad_sch

Title: Soil Moisture Meter

Size: A3	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 1/1